

1. The general case has been shown, in detail, in a worked out example. Please refer to it, for details.

(a) With the given data, critical region of the exact test is

$$w: \text{obsd } |t| > t_{3; 0.05}$$

$$\therefore \text{obsd } |t| = 6.80 \text{ and tabulated } t_{3; 0.05} = 2.353,$$

reject H_0 at 10% level of significance.

critical region of the approximate test is

$$w: \text{obsd } (-2 \log \lambda(\bar{x})) > \chi^2_{1; 0.10}$$

$$\therefore \text{obsd } (-2 \log \lambda(\bar{x})) = 11.19; \text{ Tab. } \chi^2_{1; 0.10} = 2.706,$$

reject H_0 at 10% level of significance.

(The asymptotic test matches with the exact test even for a sample size as low as 4!).

(b) With $\sigma^2 = 1$, the test statistic is, say, $W = 2(\bar{X} - 2) \sim N(2(\mu - 2), 1)$
Prob. (Type II error) = $P(\text{Accept } H_0 | \mu)$, as a fn of μ (not replaced by μ_0).
 $= P(|W| < c_{\alpha/2} | \mu) = \Phi(c_{\alpha/2} - 2\mu + 4) - \Phi(-c_{\alpha/2} - 2\mu + 4)$
 (Note the change in rejection criterion from the unknown σ^2 case).

$$\frac{2}{\sup_{\mu \in \mathbb{R}_0} L(\mu | \bar{x}) = L(\mu_0 | \bar{x}) = (2\pi\sigma^2)^{-n/2} \exp. \left\{ -\frac{1}{2\sigma^2} \sum (x_i - \mu_0)^2 \right\}}$$

$$\text{Note, } \hat{\mu}_{MLE} = \begin{cases} \bar{x} & \text{if } \bar{x} \leq \mu_0 \\ \mu_0 & > \end{cases}$$

$$\therefore \sup_{\mu \in \mathbb{R}_0 \cup \mathbb{R}_A} L(\mu | \bar{x}) = \begin{cases} (2\pi\sigma^2)^{-n/2} \exp. \left\{ -\frac{1}{2\sigma^2} \sum (x_i - \bar{x})^2 \right\} & \text{if } \bar{x} \leq \mu_0 \\ (2\pi\sigma^2)^{-n/2} \exp. \left\{ -\frac{1}{2\sigma^2} \sum (x_i - \mu_0)^2 \right\} & > \end{cases}$$

(restricted alt. case)

$$\text{i.e., } \underline{\lambda(\bar{x})} = \begin{cases} \exp. \left\{ -\frac{n}{2\sigma^2} (\bar{x} - \mu_0)^2 \right\} & \text{if } \bar{x} \leq \mu_0 \\ 1 & > \end{cases}$$

Critical region

$$w: \lambda(\bar{x}) < c \Leftrightarrow \bar{x} < k, \quad k \text{ from size-}\alpha \text{ condn, i.e.,}$$

$$P_{H_0}(\bar{X} < k) = \alpha \Rightarrow k = \mu_0 - \frac{\sigma}{\sqrt{n}} z_{\alpha}.$$

$$\left| \begin{array}{l} \text{Use } \Phi(z_{\alpha}) = 1 - \alpha \\ \text{i.e., } \Phi(1 - \alpha) = \alpha, \\ \text{but } z_{1 - \alpha} = -z_{\alpha}. \end{array} \right.$$

3.

Under H_0 , $\hat{\mu}_{MLE} = \bar{x}$ Unconditionally, $\hat{\mu}_{MLE} = \bar{x}$, $\hat{\sigma}_{MLE}^2 = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2$

$$\sup_{\mu} L(\mu, \sigma_0 | z) = (2\pi\sigma_0^2)^{-n/2} \exp\left\{-\frac{1}{2\sigma_0^2} \sum_{i=1}^n (x_i - \bar{x})^2\right\}$$

$$\sup_{\mu, \sigma} L(\mu, \sigma | z) = (2\pi\hat{\sigma}^2)^{-n/2} \exp\left\{-\frac{1}{2\hat{\sigma}^2} \sum_{i=1}^n (x_i - \bar{x})^2\right\} \quad \therefore \text{so that the}$$

$$\text{L.R. criterion } \lambda(z) = \frac{\sup_{\mu, \sigma_0} L(\mu, \sigma_0 | z)}{\sup_{\mu, \sigma} L(\mu, \sigma | z)} = \left(\frac{\hat{\sigma}^2}{\sigma_0^2}\right)^{n/2} \exp\left\{-\frac{n}{2}\left(\frac{\hat{\sigma}^2}{\sigma_0^2} + 1\right)\right\}$$

$$= \left\{ \frac{\hat{\sigma}^2}{\sigma_0^2} \exp\left(-\frac{\hat{\sigma}^2}{\sigma_0^2} + 1\right) \right\}^{n/2}, \quad \text{and the critical region}$$

$$w: \lambda(z) < c \Leftrightarrow \frac{\hat{\sigma}^2}{\sigma_0^2} \exp\left(-\frac{\hat{\sigma}^2}{\sigma_0^2} + 1\right) < k.$$

$$\Leftrightarrow \frac{\hat{\sigma}^2}{\sigma_0^2} < k_1 \quad \text{or} \quad \frac{\hat{\sigma}^2}{\sigma_0^2} > k_2$$

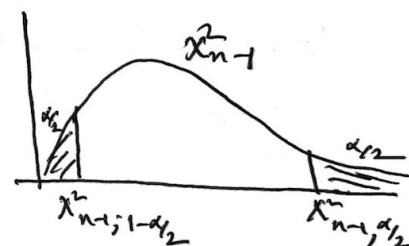
(Use distribution of $\hat{\sigma}^2$ & size condition, to obtain k_1 & k_2)

$$\text{Now, } \frac{n\hat{\sigma}^2}{\sigma_0^2} = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{\sigma_0^2} \sim \chi_{n-1}^2 \quad \text{under } H_0 \quad (\text{when } \sigma^2 = \sigma_0^2).$$

Considering a equal-tailed cut-off points,

$$P_{H_0}\left(\frac{n\hat{\sigma}^2}{\sigma_0^2} < nk_1\right) = \alpha/2 \quad \text{and} \quad P_{H_0}\left(\frac{n\hat{\sigma}^2}{\sigma_0^2} > nk_2\right) = \alpha/2, \quad \text{we get}$$

$$k_1 = \frac{1}{n} \chi_{n-1; 1-\alpha/2}^2 \quad \text{and} \quad k_2 = \frac{1}{n} \chi_{n-1; \alpha/2}^2$$



4.

$$L(\theta | z) = \prod_{i=1}^n f_{\theta}(x_i)$$

$$\sup_{\theta \in \Theta_0} L(\theta | z) = L(\theta_0 | z) = \prod_{i=1}^n f_{\theta_0}(x_i)$$

$$\sup_{\theta \in \Theta_A} L(\theta | z) = \begin{cases} L(\theta_0 | z) & \text{if } \prod_{i=1}^n f_{\theta_0}(x_i) > \prod_{i=1}^n f_{\theta_1}(x_i) \\ L(\theta_1 | z) & \text{if } \prod_{i=1}^n f_{\theta_0}(x_i) < \prod_{i=1}^n f_{\theta_1}(x_i) \end{cases}$$

$\Theta = \{\theta_0, \theta_1\}$,
with $\theta_0 \neq \theta_1$

$$\therefore \text{L.R. criterion, } \lambda(z) = \begin{cases} 1 & \text{if } \frac{\prod_{i=1}^n f_{\theta_1}(x_i)}{\prod_{i=1}^n f_{\theta_0}(x_i)} < 1 \\ \frac{L(\theta_0 | z)}{L(\theta_1 | z)} & \text{if } \frac{\prod_{i=1}^n f_{\theta_1}(x_i)}{\prod_{i=1}^n f_{\theta_0}(x_i)} > 1 \end{cases}$$

> , & the critical region,

$$w: \lambda(z) < c \Leftrightarrow \frac{\prod_{i=1}^n f_{\theta_1}(x_i)}{\prod_{i=1}^n f_{\theta_0}(x_i)} > k, \quad \text{which is the c.r. of the NP test}$$

$$L(\theta|x) = \theta^n \left(\prod_{i=1}^n x_i \right)^{\theta-1}, \quad \theta > 0; \quad 0 < x_i < 1$$

$$\ln L(\theta|x) = n \ln \theta + (\theta-1) \sum \ln x_i$$

Unrestricted mle, $\hat{\theta}_{mle} = -\frac{n}{\sum \ln x_i} = \hat{\theta}$, say

Under H_0 , $\hat{\theta}_{mle} = \begin{cases} \hat{\theta} & \text{if } \hat{\theta} > \theta_0 \\ \theta_0 & \text{if } \hat{\theta} < \theta_0 \end{cases}$

$$\therefore \sup_{\theta \in \Theta_0} L(\theta|x) = \sup_{\theta \in \Theta_0} L(\theta|x) \quad \text{if } \hat{\theta} > \theta_0 \quad \text{and} \quad \sup_{\theta \in \Theta} L(\theta|x) = \sup_{\theta \in \Theta_0} L(\theta|x) = \theta_0^n \left(\prod_{i=1}^n x_i \right)^{\theta_0-1} \quad \text{if } \hat{\theta} < \theta_0$$

$$\text{ie, } \lambda(x) = \begin{cases} 1 & \text{if } -\frac{n}{\sum \ln x_i} \geq \theta_0 \\ \left(\frac{\theta_0}{-\frac{n}{\sum \ln x_i}} \right)^n \frac{(\prod x_i)^{\theta_0-1}}{(\prod x_i)^{-(\frac{n}{\sum \ln x_i})+1}} & \text{if } -\frac{n}{\sum \ln x_i} < \theta_0 \end{cases}$$

Then, w: $\lambda(x) < c \Leftrightarrow -\frac{n}{\sum \ln x_i} < k \Leftrightarrow -\sum \ln x_i > k^*$

Now, $2\theta_0 (-\sum \ln x_i) \sim \chi^2_{2n}$ under H_0 ($\theta = \theta_0$).

and hence, $P_{H_0}(2\theta_0 (-\sum \ln x_i) > 2\theta_0 k^*) = \alpha$
 $\Rightarrow k^* = \frac{1}{2\theta_0} \chi^2_{2n; \alpha}$

The size- α LR test is given by

Rej. H_0 if $(\hat{\theta}) - \sum \ln x_i > \frac{1}{2\theta_0} \chi^2_{2n; \alpha}$.

(Check the power for behaviour to comment on whether this is equivalent to the corresponding UMP test or not)

6.

Let $Y_i = \#$ users of the product in city i . $\therefore Y \sim \text{Bin}(n_i, p_i), i=1, 2, 3$.

$H_0: p_1 = p_2 = p_3 (=p, \text{ say, unknown})$ ag. $H_A: H_0 \text{ not true}$

$\Theta_0 = \{(p_1, p_2, p_3) : 0 \leq p_1 = p_2 = p_3 \leq 1\}$, $\Theta_A = \{(p_1, p_2, p_3) : 0 \leq p_1, p_2, p_3 \leq 1\}$.

$\dim \Theta_0 = 1$

$\dim \Theta_A = 3$.

$$\lambda(x) = \frac{\sup_{\Theta_0} L(p|x)}{\sup_{\Theta_A} L(p|x)} = \frac{(\hat{p})^{y_1+y_2+y_3} (1-\hat{p})^{n-y_1-y_2-y_3}}{(\hat{p}_1)^{y_1} (1-\hat{p}_1)^{n_1-y_1} (\hat{p}_2)^{y_2} (1-\hat{p}_2)^{n_2-y_2} \hat{p}_3 (1-\hat{p}_3)^{n_3-y_3}}$$

Under H_0 , mle of $p (=p_1 = p_2 = p_3)$ is $\hat{p} = \frac{y_1+y_2+y_3}{n_1+n_2+n_3} = 0.22$

Unrestricted mle of $\hat{p}_i = \frac{y_i}{n_i}, i=1, 2, 3$; $\hat{p}_1 = 0.14, \hat{p}_2 = 0.27, \hat{p}_3 = 0.235$

(Under H_0 , we cannot ascertain the exact distn of $\lambda(x)$, hence use the (or, otherwise) asymptotic test), critical region is

w: And $(-2 \ln(\lambda(x))) > \chi^2_{2; 0.05} = 5.99$.

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Accept H_0 at 5% level.